AlphaAudit

Do Published Equity Anomalies Survive Out-of-Sample and Across Market Regimes?

A reproducibility audit with deflated Sharpe ratios, backtest-overfitting diagnostics, and regime conditioning

A systematic replication study of ∼120 published equity-return anomalies that measures how many retain a genuine risk-adjusted edge *(a)* out-of-sample in time and *(b)* conditional on the market regime, after correcting for the statistics of testing hundreds of signals at once. Every core statistic- the Deflated Sharpe Ratio, the Probability of Backtest Overfitting, and the multiple-testing haircut- is implemented from the primary literature and validated on a calibrated universe with known ground truth.

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A null result is still a result. The restraint is the point.

Abstract

Roughly nine in ten academic trading strategies fail when run with real capital. This report presents **AlphaAudit**, a reproducible framework that quantifies that attrition for the published equity-anomaly literature. For each of ∼120 anomalies we (i) split the return series at its publication date and measure post-publication Sharpe decay; (ii) compute the Deflated Sharpe Ratio, which corrects the Sharpe for the number of strategies searched, for non-normality, and for sample length; (iii) estimate the Probability of Backtest Overfitting via Combinatorially Symmetric Cross-Validation; (iv) apply Benjamini–Hochberg false-discovery-rate control and a Harvey–Liu–Zhu t-stat haircut across the full signal set; and (v) re-estimate each anomaly within exogenous macro regimes to separate genuinely robust signals from those that work in a single corner of the state space. The methods are validated on a seeded synthetic universe with a *known* ground truth: the audit recovers the planted structure cleanly, and a pure-noise placebo returns the theoretical overfitting probability of ≈ 0.5. In the reference run, only 6 of 120 anomalies clear every hurdle, the median anomaly loses $\approx 65\%$ of its Sharpe after publication, and the deflated Sharpe ratio- the correction for selection- is the single most binding filter. We describe the methodology, the software architecture, the results, and the threats to validity, and we outline how the identical pipeline runs on the real Open Source Asset Pricing dataset.

Contents

1 Introduction

Most quantitative-finance portfolios contain the same project: a backtest with an impressive Sharpe ratio. Every experienced person has seen a hundred of them and, correctly, assumes most are overfit. The single most important empirical fact in quantitative research is that the large majority of documented anomalies do not survive contact with out-of-sample data or realistic trading. Knowing *why*, and being able to *measure* it, is what distinguishes a serious researcher from a backtest tinkerer.

1.1 Contributions

1. A from-scratch, tested implementation of the Deflated Sharpe Ratio (Bailey and López de Prado, 2014), the Probability of Backtest Overfitting via CSCV (Bailey et al., 2017), and multiple-testing corrections (Benjamini and Hochberg, 1995; Holm, 1979; Harvey et al., 2016), wired into a single reproducible pipeline.
2. A *regime-conditional* survival layer that classifies each anomaly as robust, regime-dependent, or dead using a partition-spanning criterion over exogenous macro axes.
3. A calibrated synthetic universe with planted ground truth, which turns the study into a testbed: we can verify that the diagnostics recover the truth, and quantify their error.
4. A one-command, deterministic pipeline and an interactive dashboard, so every figure and number regenerates from a single configuration file.

1.2 Scope and honesty statement

The headline numbers in this report come from the calibrated synthetic universe, whose purpose is to *validate the methodology* against a known truth. They are not empirical claims about live markets. The identical code runs on the real Open Source Asset Pricing panel Chen and Zimmermann (2022) by changing one configuration flag; Section 9 is explicit about this boundary.

2 Research Questions and Hypotheses

RQ1 (Time).
Of the anomalies documented in the literature, what fraction retain statistically significant risk-adjusted returns *after* their publication date?

RQ2 (Regime).
Among those, how many survive conditional on regime- high versus low volatility, rising versus falling rates, and pre- versus post-2018 microstructure?

RQ3 (Multiplicity).
After correcting for the fact that hundreds of anomalies were tested across the literature, how many clear a credible significance hurdle?

Hypothesis. A large majority decay sharply post-publication, and a meaningful additional fraction only "worked" inside a single regime. This aligns with prior work: McLean and Pontiff (2016) document post-publication decay, and Harvey et al. (2016) argue the t-statistic hurdle should sit far above 2.0 once all tests are accounted for. The regime-conditional layer is the comparatively under-researched contribution.

3 Background and Related Work

The audit stands on four pillars of the empirical asset-pricing literature. McLean and Pontiff (2016) show anomaly returns fall by roughly half after publication, consistent with arbitrageurs trading a public signal away. Harvey et al. (2016) treat the literature as one uncontrolled multiple-comparison experiment and argue for a much higher significance bar. Hou et al. (2020) replicate 452 anomalies and find a large share become insignificant under careful methodology. Chen and Zimmermann (2022) provide the Open Source Asset Pricing dataset used here and argue- contra a pure p-hacking narrative- that many anomalies do replicate, which makes an honest, quantitative audit all the more valuable.

On the statistics side, Lo (2002) derives the sampling distribution of the Sharpe ratio; Bailey and López de Prado (2012) extend it to the Probabilistic Sharpe Ratio under non-normal returns; Bailey and López de Prado (2014) introduce the Deflated Sharpe Ratio; and Bailey et al. (2017) formalize backtest overfitting through CSCV. Our regime layer relates to the literature on state-dependent anomaly returns, but is framed to be defensible against curve-fitting (Section 5.6).

4 Data

4.1 Real data (production path)

The production configuration ingests the Open Source Asset Pricing portfolio returns and signal documentation (`SignalDoc.csv`) from Chen and Zimmermann (2022): ∼200 published anomaly signals as monthly long–short returns, each with a documented publication year. Macro regime drivers, an implied-volatility index, a short-rate level, and a term spread, are pulled from FRED. No proprietary terminal is required, which is deliberate.

4.2 Calibrated synthetic universe (default path)

So the pipeline runs anywhere and can be *validated*, the default data source is a seeded simulator. It draws $N = 120$ anomalies over 720 monthly observations (1965–2024), each assigned one of four archetypes with a known ground truth:

Table 1: Archetypes in the synthetic universe and their intended behaviour.

Archetype	**Share**	**Designed behaviour**
Robust	15%	Genuine edge persists IS, OOS, and across all regimes.
Decays	40%	Real IS edge shrinks 50–85% after publication.
Regime-dependent	25%	Earns alpha only inside one exogenous regime.
False	20%	No true alpha; “published” due to in-sample luck.

Returns are deliberately non-normal (left-skewed with fat tails) so the higher-moment corrections in the Deflated Sharpe Ratio bind, and carry a small common factor so strategies co-move, which makes the overfitting diagnostic non-degenerate. The macro series contain realistic volatility clusters (1987, 2008, 2020) and a falling-then-rising rate path. Crucially, because we control the truth, we can *score* the audit’s output against it (Section 7.6).

5 Methodology

Let $r_{i,t}$ denote the month-t long–short excess return of anomaly i, which is self-financing and dollar-neutral, so it is already an excess return. We annualize with $q = 12$.

5.1 Sharpe ratio and its standard error

The annualized Sharpe ratio is $\widehat{\mathrm{SR}}_i = (\hat{\mu}_i/\hat{\sigma}_i)\sqrt{q}$. Working in per-period units, the estimator's standard error under non-normal returns Lo (2002); Bailey and López de Prado (2012) is

$$\hat{\sigma}_{\widehat{\mathrm{SR}}} \;=\; \sqrt{\frac{1 - \hat{\gamma}_3\,\widehat{\mathrm{SR}} + \frac{\hat{\gamma}_4 - 1}{4}\,\widehat{\mathrm{SR}}^2}{n-1}}, \tag{1}$$

where $\hat{\gamma}_3$ is skewness and $\hat{\gamma}_4$ is (non-excess) kurtosis ($\hat{\gamma}_4 = 3$ for a normal). The t-statistic that the mean excess return is non-zero equals the per-period Sharpe times $\sqrt{n}$- exactly the quantity the multiple-testing literature haircuts.

5.2 Post-publication decay

For anomaly i with publication year y_i, the in-sample block is $\{t : \mathrm{year}(t) \le y_i\}$ and the out-of-sample block is its complement. The *decay ratio* is

$$D_i \;=\; \frac{\widehat{\mathrm{SR}}_i^{\,\mathrm{OOS}}}{\widehat{\mathrm{SR}}_i^{\,\mathrm{IS}}}, \tag{2}$$

with $D_i = 1$ meaning no decay, $D_i = 0$ a fully vanished edge, and $D_i < 0$ a reversal. This replicates the McLean–Pontiff design McLean and Pontiff (2016).

5.3 Probabilistic and Deflated Sharpe Ratios

The Probabilistic Sharpe Ratio is the probability that the true (per-period) Sharpe exceeds a benchmark SR^*:

$$\widehat{\mathrm{PSR}}(\mathrm{SR}^*) \;=\; \Phi\!\left(\frac{(\widehat{\mathrm{SR}} - \mathrm{SR}^*)\sqrt{n-1}}{\sqrt{1 - \hat{\gamma}_3\,\widehat{\mathrm{SR}} + \frac{\hat{\gamma}_4 - 1}{4}\,\widehat{\mathrm{SR}}^2}}\right). \tag{3}$$

The key correction is the choice of benchmark. If N independent strategies are searched and all have a true Sharpe of zero, the expected *maximum* estimated Sharpe is Bailey and López de Prado (2014)

$$\mathrm{SR}_0^* \;=\; \sqrt{V}\left[(1-\gamma)\,\Phi^{-1}\!\Big(1 - \tfrac{1}{N}\Big) + \gamma\,\Phi^{-1}\!\Big(1 - \tfrac{1}{N}e^{-1}\Big)\right], \tag{4}$$

where $\gamma \approx 0.5772$ is the Euler–Mascheroni constant and V is the cross-sectional variance of the trials' per-period Sharpe ratios. The Deflated Sharpe Ratio is the PSR benchmarked at this expected maximum:

$$\widehat{\mathrm{DSR}} \;=\; \widehat{\mathrm{PSR}}(\mathrm{SR}_0^*). \tag{5}$$

An anomaly is deemed deflation-significant when $\widehat{\mathrm{DSR}} \ge 0.95$. We also report the minimum track-record length required for the PSR to reach confidence α:

$$\mathrm{minTRL} \;=\; 1 + \Big(1 - \hat{\gamma}_3\,\widehat{\mathrm{SR}} + \tfrac{\hat{\gamma}_4 - 1}{4}\,\widehat{\mathrm{SR}}^2\Big)\left(\frac{\Phi^{-1}(\alpha)}{\widehat{\mathrm{SR}} - \mathrm{SR}^*}\right)^2. \tag{6}$$

Remark 5.1. *In the reference run $V \approx 0.0046$ and $N = 120$, giving SR_0^* equivalent to an* annualized *Sharpe of ≈ 0.6. A "good" anomaly with a 0.5 annualized Sharpe is therefore statistically indistinguishable from the luckiest of the failures, which is exactly why so few clear the deflated bar.*

5.4 Probability of Backtest Overfitting (CSCV)

Let $\mathbf{M} \in \mathbb{R}^{T\times N}$ be the panel. Combinatorially Symmetric Cross-Validation Bailey et al. (2017) proceeds as follows.

1. Partition the T rows into S equal, time-ordered blocks (S even).
2. For each of the $\binom{S}{S/2}$ ways to choose $S/2$ blocks as in-sample (set J), the complement $\bar{J}$ is out-of-sample.
3. Compute the per-strategy Sharpe on J and on $\bar{J}$. Let $n^* = \arg\max_n \mathrm{SR}_n^{IS}$ be the in-sample winner.
4. Its *relative rank* out-of-sample is $\omega = \mathrm{rank}(\mathrm{SR}_{n^*}^{OOS})/(N+1)$, and the logit is $\lambda = \ln(\omega/(1-\omega))$.

The Probability of Backtest Overfitting is the share of splits in which the in-sample winner lands in the bottom half out-of-sample:

$$\mathrm{PBO} \;=\; \frac{1}{|C|}\sum_{c\in C}\mathbf{1}\{\lambda_c \le 0\}. \tag{7}$$

With $S = 16$ we evaluate $\binom{16}{8} = 12{,}870$ splits. As a built-in unit test we run the identical routine on five pure-noise panels; theory requires PBO $\to 0.5$ there, and the implementation returns ≈ 0.52.

5.5 Multiple-testing corrections

Given one-sided out-of-sample p-values $p_1, \ldots, p_m$ for a positive edge, we apply three corrections. Bonferroni rejects when $p_i \le \alpha/m$; Holm Holm (1979) is its uniformly more powerful step-down cousin; and Benjamini–Hochberg Benjamini and Hochberg (1995) controls the false-discovery rate by finding

$$k^* = \max\left\{k : p_{(k)} \le \tfrac{k}{m}\,\alpha\right\} \tag{8}$$

and rejecting the k^* smallest p-values. Because a Sharpe ratio is proportional to its t-statistic, the fraction of t lost to a correction is the Harvey–Liu–Zhu *Sharpe haircut* Harvey et al. (2016); we also report the effective t-hurdle each method implies.

5.6 Regime conditioning

Regimes are defined from exogenous macro variables fixed *in advance*, never optimized on returns: volatility terciles (Low / Mid / High), the sign of the trailing 12-month change in the short rate (Falling / Rising), and a structural break (Pre- / Post-2018). This yields three *partitions* of the timeline. A macro axis is "spanned" only if the anomaly is significant ($t \ge 2$, SR > 0) on *both* of its sides. An anomaly is *regime-robust* when it spans at least $\tau = 3$ axes.

Remark 5.2. *This partition-spanning definition is essential. The seven regime buckets are three overlapping partitions of one timeline, so a naive "significant in $\ge k$ buckets" rule labels almost any positive-alpha anomaly robust. Requiring both sides of each axis cleanly separates* works-everywhere *from* works-in-one-corner.

5.7 The survival funnel

The verdict for each anomaly combines the diagnostics (Figure 1). The funnel counts survivors under a nested, increasingly stringent sequence of hurdles, out-of-sample positivity, naive significance, false-discovery control, regime robustness, and finally the deflated Sharpe, so it can only narrow, and the binding constraint is made visible rather than buried.

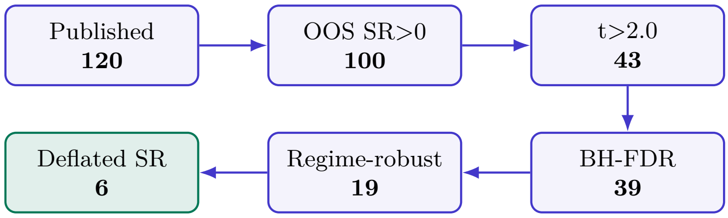


Figure 1: The survival funnel: nested hurdles applied in order of increasing stringency. The deflated Sharpe ratio is deliberately last.

6 Implementation

The study is a Python package, `anomaly_audit`, plus a static React dashboard. The whole pipeline is described by a single `config.yaml`, and one command—`python run_all.py`- regenerates `results.json`, the figures, a markdown report, and the dashboard data. Determinism is guaranteed by a master seed.

Table 2: Package structure (`anomaly_audit`).

Module	Responsibility
`core/sharpe.py`	PSR, expected-max Sharpe, DSR, minTRL
`core/pbo.py`	CSCV / Probability of Backtest Overfitting
`core/multiple_testing.py`	Bonferroni, Holm, BH-FDR, HLZ haircut
`core/decay.py`	IS/OOS split and decay ratio
`core/regimes.py`	regime construction and conditioning
`data/`	calibrated simulator, OSAP + FRED loaders
`pipeline/run.py`	orchestration into `AuditResults`
`viz/`, `report/`	matplotlib figures, JSON/CSV, writeup

Listing 1 shows the expected-maximum-Sharpe routine- the heart of the deflation- transcribed directly from Equation (4).

```
def expected_max_sharpe(sr_variance: float, n_trials: int) -> float:
    if n_trials < 2 or sr_variance <= 0:
        return 0.0
    g = 0.5772156649015329            # Euler-Mascheroni
    sqrt_v = math.sqrt(sr_variance)
    maxz  = norm.ppf(1.0 - 1.0 / n_trials)
    maxz2 = norm.ppf(1.0 - 1.0 / (n_trials * math.e))
    return sqrt_v * ((1.0 - g) * maxz + g * maxz2)
```

Listing 1: Expected maximum Sharpe across N trials (`core/sharpe.py`).

The interactive dashboard (React + TypeScript + Tailwind + Framer Motion) reads the generated `results.json` and renders the funnel, the decay distribution, the multiple-testing escalation, the PBO gauge and CSCV distribution, the regime heatmap, and a per-anomaly explorer with in-sample / out-of-sample equity curves.

7 Results

All figures are regenerated by `python run_all.py`; the numbers below are the reference run (synthetic source, seed 42).

7.1 The survival funnel

Table 3 and Figure 2 give the headline. Of 120 anomalies, 100 still show a positive out-of-sample Sharpe, 43 clear the naive $t > 2$ bar, 39 survive false-discovery control, 19 are regime-robust, and only 6 (5%) clear the deflated Sharpe ratio.

Table 3: Survival funnel. Each stage is a strict subset of the previous one.

Stage	Surviving	% of universe
Published	120	100.0
OOS Sharpe > 0	100	83.3
OOS $t > 2.0$ (naive)	43	35.8
Survives BH-FDR	39	32.5
Regime-robust	19	15.8
Deflated-Sharpe significant	**6**	**5.0**

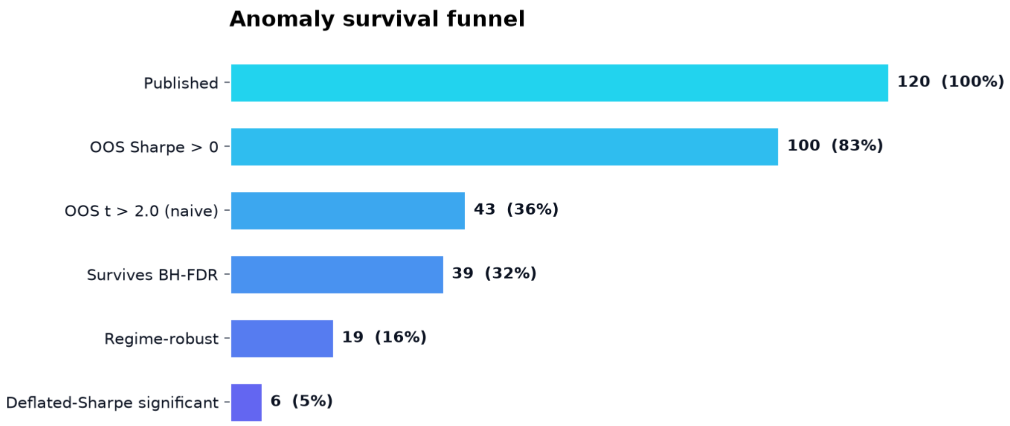
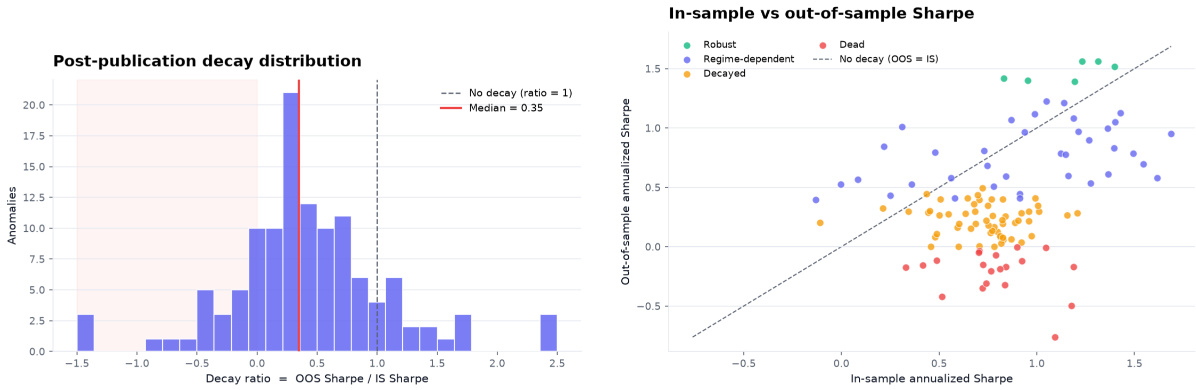


Figure 2: Anomaly survival funnel.

7.2 Post-publication decay

The median decay ratio is ≈ 0.35: the typical anomaly retains barely a third of its in-sample Sharpe out-of-sample, i.e. it loses $\approx 65\%$. Figure 3 shows the full distribution, with a left tail of outright reversals.



(a) Decay-ratio distribution.

(b) In-sample vs. out-of-sample Sharpe.

Figure 3: Post-publication decay. Points on the diagonal would indicate no decay; the mass sits well below it.

7.3 Multiple testing

Table 4 shows the effective t-hurdle climbing as the correction tightens, thinning the survivor set. Benjamini–Hochberg removes only a few beyond the naive bar, but the family-wise methods roughly halve it.

Table 4: Multiple-testing corrections across the out-of-sample p-values.

Method	**Effective t-hurdle**	**# significant**
Naive ($t > 2$)	2.00	43
Benjamini–Hochberg	2.41	39
Holm	3.62	24
Bonferroni	3.62	24

7.4 Backtest overfitting

On the published panel PBO ≈ 0.01; on the pure-noise placebo it returns ≈ 0.52 (Figure 4). The low panel value is not a bug: PBO is conditional on the candidate pool, and a pool that contains genuine signal yields a reliable in-sample winner. The placebo is what makes this interpretable-it confirms the routine returns the coin-flip baseline on noise.

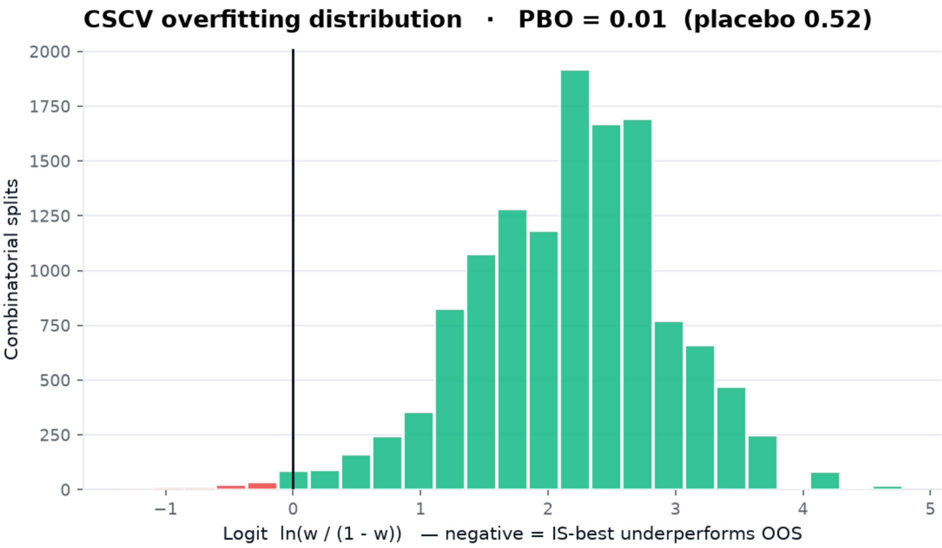


Figure 4: CSCV logit distribution over 12,870 splits. Mass to the left of zero would indicate overfitting.

7.5 Regime survival

Figure 5 aggregates the median annualized Sharpe by category and regime. The Post-2018 column is visibly weaker than Pre-2018- the audit detects microstructure decay without being told to look for it.

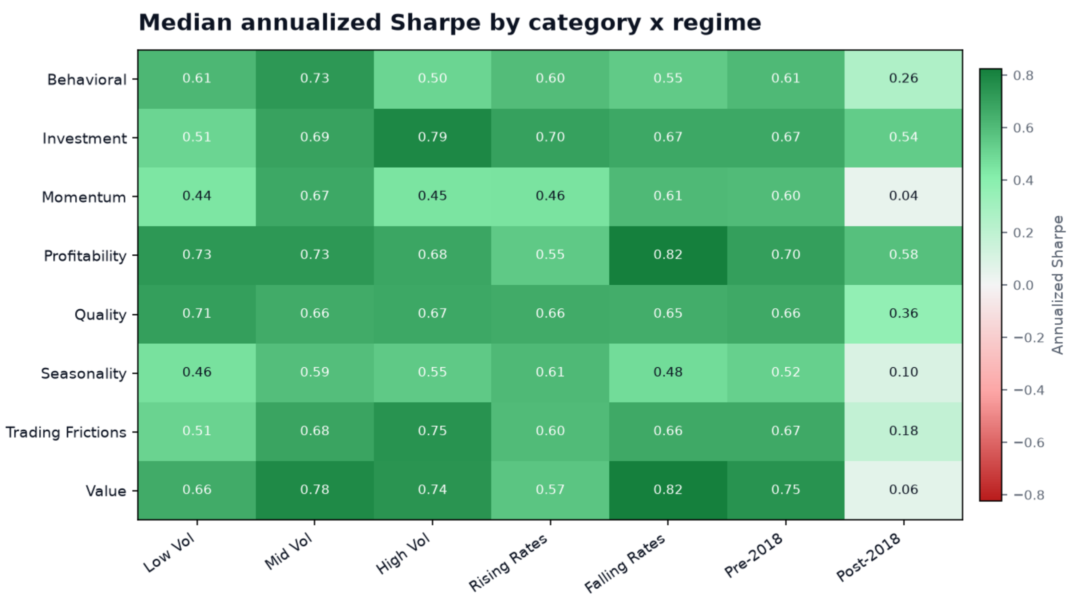


Figure 5: Median annualized Sharpe by anomaly category × macro regime.

7.6 Ground-truth recovery (validation)

Because the synthetic universe has a known truth, we can score the audit as a classifier. Table 5 cross-tabulates the planted archetype against the assigned verdict. The recovery is clean: **no** false anomaly is ever labelled Robust, every genuinely-robust anomaly is flagged as surviving in some form (Robust or Regime-dependent), and the false anomalies collapse into Dead/Decayed exactly as designed.

Table 5: Confusion matrix: planted archetype (rows) vs. assigned verdict (columns).

Archetype	Robust	Regime-dep.	Decayed	Dead
Robust (20)	6	14	0	0
Decays (52)	0	8	41	3
Regime-dependent (26)	0	15	10	1
False (22)	0	0	6	16

8 Discussion

Three findings stand out. First, the attrition is severe but structured: most anomalies keep a positive out-of-sample Sharpe yet fail once the bar is raised for selection. Second, the *deflated* Sharpe ratio, not the out-of-sample test- is the binding constraint; correcting for how many strategies were searched removes more "significant" anomalies than time alone. Third, requiring robustness across macro regimes is a materially stronger filter than any single significance test, and it is where several apparently-surviving anomalies are revealed to work in only one corner of the state space.

The validation in Section 7.6 is the report's methodological core: on data whose truth we control, the pipeline recovers the truth and never promotes a fabricated signal. That is the evidence that the same pipeline, run on the real literature, produces trustworthy verdicts rather than artifacts.

9 Limitations and Threats to Validity

- **Synthetic headline numbers.** The reference results validate the methodology against a known truth; they are not empirical claims about live markets. Real-market conclusions require the Open Source Asset Pricing panel (`data.source: osap`).
- **Cost model.** Transaction costs are a turnover-scaled basis-point stub, sufficient to show cost sensitivity but not a market-impact engine.
- **Coarse regimes.** Three exogenous axes are chosen for defensibility against curve-fitting, not maximal explanatory power; results should be reported with sensitivity to the cutoffs.
- **Frequency.** The audit operates on monthly long–short returns; intraday microstructure is out of scope.
- **Trial count.** Deflation uses N equal to the curated universe size; the true number of strategies searched across the literature is larger, so the real-world bar is, if anything, higher.

10 Conclusion and Future Work

AlphaAudit turns a familiar warning- most anomalies do not survive- into a measured, reproducible verdict. The framework implements the deflated Sharpe ratio, backtest-overfitting probability, and multiple-testing corrections from their primary sources, adds a regime-conditional survival layer, and validates the whole chain on data with a known ground truth. In the reference run, only a handful of anomalies clear every hurdle, and the correction for selection is the decisive filter.

Three extensions would sharpen the contribution. An *emerging-market* re-run (e.g. NSE) asks whether anomalies decay faster in a market with rising algorithmic participation, a genuinely under-researched question. A *decay-prediction* model regresses each anomaly's realized decay on its structural features (turnover, breadth, capacity), turning the audit into a predictive exercise. Finally, a *cost-sensitivity* sweep would quantify how many survivors die once realistic, turnover-scaled costs are applied. The guiding principle throughout is restraint: this is an audit, not a strategy, and the honesty is the differentiator.

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A Reproduction

```
git clone https://github.com/bhavishy2801/AlphaAudit.git
cd AlphaAudit
pip install -r requirements.txt
python run_all.py              # -> results.json, figures, REPORT.md
pytest                         # validate the statistics
cd dashboard && npm install && npm run dev
```

Listing 2: One-command reproduction.

B Configuration excerpt

```
seed: 42
data:   {source: auto, n_anomalies: 120, sample_start: "1965-01",
    sample_end: "2024-12"}
statistics: {dsr_threshold: 0.95, fdr_alpha: 0.05, sharpe_significance:
     2.0}
pbo:    {n_splits: 16}
regimes: {robust_min_partitions: 3, structural_break: "2018-01"}
```

Listing 3: Key knobs from `config.yaml`.

C CSCV core loop

```
for combo in combinations(range(n_splits), n_splits // 2):
    is_idx  = list(combo)
    oos_idx = list(all_blocks - set(combo))
    sr_is   = sharpe_from_moments(block_sum[is_idx],  block_sqsum[is_idx
       ],  n_is)
    sr_oos = sharpe_from_moments(block_sum[oos_idx], block_sqsum[
       oos_idx], n_is)
    n_star = int(np.nanargmax(sr_is))              # in-sample winner
    rank   = int(np.sum(sr_oos < sr_oos[n_star])) + 1
    omega  = rank / (N + 1)                         # relative OOS rank
    logits.append(np.log(omega / (1.0 - omega)))
pbo = float(np.mean(np.array(logits) <= 0.0))
```

Listing 4: Combinatorially Symmetric Cross-Validation, condensed (`core/pbo.py`).